

LatticeRank Phase 2 — Failure Analysis

Two kinds of number appear below and they must not be confused. **\$1 is the only evaluation on organizer-generated data:** the 20-pair Applied Materials sample set, run once, blind, after the solver was frozen, with nothing tuned on it. Everything from \$3 onward is measured on **our own generator**, which we built deliberately harder than the spec — it renders the reference and the search as independent acquisitions, where the organizers cut the reference from the search canvas. Those internal figures are a robustness stress-test, not a forecast, and they read roughly 30% where the official sample reads 98%.

1. Official sample: 38.8 / 40 localization, 79.1 / 85 on the scored blocks

results/phase2_experiments/official_sample_evaluation.json. 20 pairs: 8 Set A nominal, 6 Set B degraded (severity 1–4), 4 Set C absent, 2 Set D RGB; 9 architecture presets; zoom spans 8.00–12.00 at both endpoints; rotation –4.9 to +4.9°. Scored with the organizers' own rubric, including their rule that localization credit is zeroed when *found*=0. **Zero failures, zero missing rows, relative paths in pairs.csv resolved correctly from an unrelated working directory.**

Block	Result	Score	Detail
Localization	A 0.975 · B 0.967 · D 1.000	38.8 / 40	every present pair localized; misses are 1–2 px tier slips, never a wrong site
Pose (gated)	mean credit 0.866	17.3 / 20	third refinement pass added 1.7 pts; residual misses are near-ties on the tiers
Rejection	TP 16, FP 2, FN 0 → F1 0.9412	14.1 / 15	clears the F1 ≥ 0.90 gate: +4 bonus
Confidence	AUC 0.8889 vs correctness	8.9 / 10	score column, not the found flag
RGB bonus	Set D 0.900, A–C 0.982	+6 unlocked	requires D ≥ 0.40 and A–C ≥ 0.50
Runtime	median 2.98 s, max 3.24 s	inside budget	budget 5 s median; hard cap 20 s never approached

For calibration, the organizers' README reports their own naive ZNCC baseline at **0.800** mean credit on the same present pairs, with Set A “too easy” and severity 3–4 defeating it outright (p011, p012, p014 → 0). LatticeRank scores 1.00 on each of those three.

2. Where the remaining points are

Pose (4.4 pts). Every shortfall is a near-miss: p005 scale 2.33% / 0.57°, p007 1.07% / 0.68°, p003 0.32% / 0.82°, p019 1.65% / 0.60°. Full credit needs ≤1% and ≤0.25°; the coarse grid is 0.5 in scale and 3° in rotation, and their README states plainly that “a finer search or peak interpolation is required to earn top marks — which is the intended incentive.” A local fine search at the already-chosen site cannot move the coordinate, so it cannot cost localization. **Rejection (1.3 pts).** The presence model reports *found*=1 on three absent pairs (p015, p016, p018). The *score* column separates present from absent perfectly on this set — every absent site (max 0.0769) is below every present score (min 0.1064). This is recorded, not acted on: choosing a threshold from it would be tuning on organizer data, which the addendum lists as a no-appeal disqualification. **Runtime.** *dense_pose_search* is 75.6% of the 5.66 s mean (45 full-resolution correlations); presence features 12.4%, refinement 9.4%. The Deadline guard sheds stages before the 20 s cap.

3. Why our own corpus reads 30%, and why that is the right stress test

Our generator (data/phase2/p2_val + p2_stress) scores **29.6%** ≤1 px at n=280, and we measured *why* rather than assuming. Decoys and the severity ladder — the two things it was built to stress — have **no effect** (30.0 / 31.8 / 30.0% across severity 0–1 without decoys, with decoys, and the full mix; *set_a_calibration.json*). What does have an effect is that we render the reference and the search as **independent acquisitions**, with separate noise realisations. Cropping the reference from the search pixels instead, on identical pairs with the identical pipeline, lifts ≤1 px from 30.0% to **65.0%** (paired McNemar 26 v 5, p = 1.9e–4; *samecanvas_bound.json*). The organizers' README confirms that their reference is cut from the search canvas — which is why the official sample reads 98% and ours reads 30%. The corpus is not mis-built; it is a harder problem than the one being scored, and it is the reason we know the shipped preprocessing is at a local optimum rather than a lucky default.

That optimum was established by exhaustion, all paired on identical pairs, all end-to-end top-1 accuracy: an oracle handed the ground-truth pose still selects the true site only 31.2% of the time (*localization_ceiling.json*), so search is not the bottleneck; cross-pose sum / votes / peakiness score 2.5 / 1.2 / 18.8% against 25.0%; dense periodic residual is indistinguishable (p = 1.00); five shortlist ranking rules are all below peak; trimmed ZNCC degrades *monotonically* as pixels are dropped, the signature of uniform independent noise rather than occlusion; a 12-configuration DoG band sweep leaves the shipped (2, 8) best and collapses to 0% above the lattice pitch, because the lattice is what lets the correlation lock a position at all; normalised-gradient-field re-ranking degrades monotonically with its weight (13.3% pure); and a nearest-to-centre tie-break loses even on the slice selected so that the convention holds (51.4% → 45.7%, *centre_tiebreak_natural.json*). Ten independent negatives converge on one mechanism.

4. What worked: DoG band-pass, n=280

results/phase2_experiments/exp01/summary.json, identical pose sweep and selection rule, only the ZNCC input differing: raw 18.6% → DoG **29.6%**, credit 0.189 → 0.334; paired fixed 49 / broke 18, net +31, McNemar **p = 0.0002**. The gain concentrates at severity 3 (+22.0 pp) — low-frequency charging-drift suppression, not generic contrast — which is exactly where Set B's 0.55 weight sits.

5. Final shipped architecture

Phase 1's periodic-aware ZNCC pipeline, **extended, not replaced**: dense full-resolution pose sweep (45 poses) over the disclosed ranges, DoG-band-passed ZNCC as the matching signal, refinement to subpixel / sub-degree credit, reported pose clamped to the disclosed [8, 12] and ±5° (explicitly permitted), and **two separate fitted models**: a 17-feature HGB presence model for *found* and a 20-feature logistic correctness model for *score*. They answer different questions — *is the reference here versus is the coordinate I am reporting correct* — and a present-but-mislocalized pair must score low on the second while being 1 on the first.

6. Honest limitations

- **n = 20 on the official sample.** Set B is six pairs; one miss moves the block by ±3.7 points. The 38.8 is a strong point estimate, not a guarantee on 200 pairs.
- **Runtime is inside budget:** median 2.98 s, max 3.24 s on the official set and 2.92 s on 60 of our own, 0 pairs over 5 s. Three memoizations got it there, each verified byte-identical: the anti-alias depends on scale not rotation; *scene_features* was re-refining what the entry point had already refined; and the ZNCC denominator's windowed-variance term depends on template *shape* only, so the sweep needs 63 FFT convolutions rather than 135.
- **Two false positives on absent pairs** cost 0.9 rejection points. The fix is visible in our own score column and was deliberately not taken, per the tuning rule.
- **Confidence is the weakest block under severity.** On 40 further pairs drawn off-grid from the organizers' own generator, with all 12 presets and severity weighted to 3–4, localization holds at 35.2/40 and rejection improves to F1 0.9697, but confidence AUC falls from 0.889 to 0.775. That is where this submission is softest.
- **Pose is gated on localization**, so its 20 points are a multiplier on site selection, not an independent block.
- All internal evidence is synthetic and self-generated, and is harder than the scored task; read it as a bound on robustness, never as a prediction.

7. Reproducibility

Python 3.11, CPU-only, no network at run time (scripts/verify_offline.py, audit-hook based). Weights ship inside the package (presence_hgb.pkl, correctness_lr.pkl, hgb_r2.joblib, SHA256-pinned in their metadata files) and load script-relative. scripts/build_submission.py packages from an explicit allow-list, fails if any required file is absent or if this document cites an evidence file the zip does not contain, then extracts the archive to a scratch directory and runs the entry point inside it. A 21-clause output-contract audit passes against the built zip under 3.11. Determinism: byte-identical predictions across repeated runs; 12 generator gates in scripts/validate_phase2.py.